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# INTERNATIONAL A-LEVEL

# MATHEMATICS

# MA03

(9660/MA03) Unit P2 − Pure Mathematics

Mark scheme

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*201xma03/MS*

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**Key to mark scheme abbreviations**

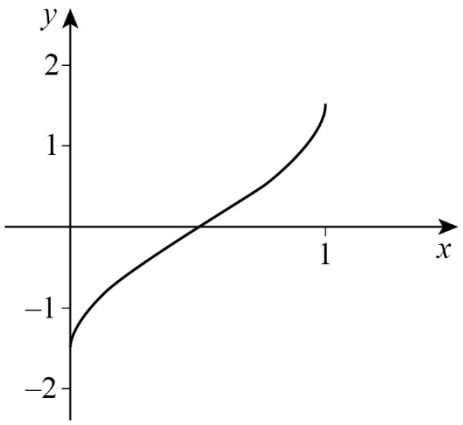
| | |
|---|---|
| **M** | Mark is for method |
| **m** | Mark is dependent on one or more M marks and is for method |
| **A** | Mark is dependent on M or m marks and is for accuracy |
| **B** | Mark is independent of M or m marks and is for method and accuracy |
| **E** | Mark is for explanation |
| **√or ft** | Follow through from previous incorrect result |
| **CAO** | Correct answer only |
| **CSO** | Correct solution only |
| **AWFW** | Anything which falls within |
| **AWRT** | Anything which rounds to |
| **ACF** | Any correct form |
| **AG** | Answer given |
| **SC** | Special case |
| **oe** | Or equivalent |
| **A2, 1** | 2 or 1 (or 0) accuracy marks |
| **–*x* EE** | Deduct *x* marks for each error |
| **NMS** | No method shown |
| **PI** | Possibly implied |
| **SCA** | Substantially correct approach |
| **sf** | Significant figure(s) |
| **dp** | Decimal place(s) |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 1(a)(i) | $[5\sin\theta - 12\cos\theta =]$<br>$R\sin\theta\cos\alpha - R\cos\theta\sin\alpha$<br>$R = 13$<br>$\alpha = 67.4$ | **M1**<br><br>**A1**<br>**A1** | PI |
| 1(a)(ii) | $\sin(\theta - 67.4) = -\frac{1}{13}$<br>$\theta = -108.2$<br>$\theta = 63.0$ | **M1**<br><br>**A1**<br>**A1** | <br><br>**Ft** *their* **(a)**<br>Both correct and no extras in interval (ignore answers outside interval) |
| 1(b) | $2\cot^2 x = 2\text{cosec}^2 x - 2 \quad [= 10 - 5\text{cosec}x]$<br>$[2\text{cosec}^2 x - 2 = 10 - 5\text{cosec}x]$<br>$2\text{cosec}^2 x + 5\text{cosec}x - 12 = 0$<br>$(2\text{cosec}x - 3)(\text{cosec}x + 4)[= 0]$<br>$\text{cosec}x = \frac{3}{2}, -4$<br>$\sin x = \frac{2}{3}, -0.25$<br>$x = 42, 138,$<br>$-14, 194$ | **M1**<br><br><br>**m1**<br><br>**A1**<br><br><br>**B1**<br>**B1** | Correct use of trig identity PI<br><br><br>Factorisation or correct use of formula<br><br>Both correct and no errors seen<br><br>Either may be seen<br><br>Sight of **any** of these values correct<br>All four correct and no extras in interval (ignore answers outside interval) |
| | **Total** | **11** | |

**Notes:**

**(a)** May use cos and sin leading to $\sin p = 2/3, -1/4$ for first **M1, m1, A1**

**(b)** Condone more accurate correct answers, but not –14.5, 194.5

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 2(a) | $0 \le x \le 1$ | B1 | |
| 2(b) |  | M1<br>A1 | Correct shape and position<br>$(0,-\frac{\pi}{2})$ $(1,\frac{\pi}{2})$<br>stated or marked on diagram |
| 2(c) | Stretch + either I or II<br>Parallel to $x$-axis I<br>SF $\frac{1}{2}$ II<br>Followed by<br>Translation $\begin{bmatrix} k \\ 0 \end{bmatrix}$<br>$k = \frac{1}{2}$ | M1<br>A1<br>M1<br>A1 | **Alt:** Translation $\begin{bmatrix} k \\ 0 \end{bmatrix}$ **M1**<br>$k = 1$ **A1**<br>Followed by<br>Stretch in $x$-direction **M1**<br>SF $\frac{1}{2}$ **A1** |
| 2(d) | $f(x) = \sin^{-1}(2x-1) + x - 1$<br>$f(0.6) = -0.198...$<br>$f(0.7) = 0.111...$<br><br>Change of sign, (the function is continuous), $0.6 < \alpha < 0.7$ | M1<br><br>A1 | Or reverse<br>Both values rounded or truncated to at least 1sf<br><br>Must have both statements and interval in words or symbols<br>Accept $x$ for $\alpha$ |
| 2(e) | $x_2 = 0.695$<br>$x_3 = 0.650$ | B1<br>B1 | |
| | **Total** | **11** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 3(a) | $u = \ln x, \quad dv = x$<br>$du = \frac{1}{x}, \quad v = \frac{x^2}{2}$<br>$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \times \frac{1}{x} \mathrm{d}x$<br>$= \frac{x^2}{2} \ln x - \frac{x^2}{4} \quad [+c]$ | M1<br>A1<br>A1 | Use of parts formula |
| 3(b) | $u = \ln x, \quad dv = 1$<br>$du = \frac{1}{x}, \quad v = x$<br>$\int \ln x \, dx = x \ln x - \int x \times \frac{1}{x} \mathrm{d}x$<br>$= x \ln x - x \quad [+c]$ | M1<br>A1<br>A1 | Use of parts formula |
| | **Total** | **6** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 4(a) | $8[(1.5)^3]+b[(1.5)^2]+c[1.5]+6=-3.75$<br>$8[(0.5)^3]+b[(0.5)^2]+c[0.5]+6=5.25$ | M1 | One correct substitution<br>OR for **M1** use of long division |
| | $2.25b+1.5c=-36.75$<br>$0.25b+0.5c=-1.75$ | A1 | |
| | $b=-21$ | m1 | Attempt to solve |
| | $c=7$ | A1 | Both answers |
| 4(b) | $(4x^2-1)(3x-2)$<br>$=12x^3-8x^2-3x+2$ | B1<br>B1 | |
| | $(12x^3-8x^2+x+7)-(12x^3-8x^2-3x+2)$<br>$=4x+5$ | M1<br>A1 | |
| | **OR** | | |
| | $12x^3-8x^2+x+7=(4x^2-1)(3x+d+\frac{ex+f}{4x^2-1})$ | (M1) | Accept other correct approaches |
| | $4d=-8,\ \ d=-2$<br>$-3+e=1,\ \ e=4$<br>$7=f-d,\ \ f=5$ | (B1)<br>(B1)<br>(A1) | |
| | **Total** | **8** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 5(a) | $\frac{12}{(3-u)(3+u)} = \frac{A}{3-u} + \frac{B}{3+u}$ oe<br>$A = 2,\ \ B = 2$ | M1<br>A1 | |
| 5(b) | $[\frac{\mathrm{d}u}{\mathrm{d}x} =]\ \cos x$ | B1 | |
| | $[\int \frac{12\cos x}{8+\cos^2 x}\mathrm{d}x =] \int \frac{12\mathrm{d}u}{8+1-u^2}$<br>$= \int \frac{12\,\mathrm{d}u}{(3-u)(3+u)}$ | M1 | All in terms of $u$, condone omission of $\mathrm{d}u$ |
| | $= \int \frac{A}{3-u} + \frac{B}{3+u}\mathrm{d}u$ | A1 | Must see $\mathrm{d}u$ here, or earlier |
| | $[\int\ \ =] -2\ln(3-u) + 2\ln(3+u)$<br>$= 2\ln\frac{3+u}{3-u}$ | M1 | Correct integration |
| | $[x]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = [u]_{0.5}^{1}$<br>$[\int\ \ =]\ 2\ln 2 - 2\ln 1.4$ | B1 | Change of limits, maybe seen earlier (may change back to $x$ and not change limits) |
| | $= \ln\frac{100}{49}$ | A1 | |
| | **Total** | **8** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 6(a)(i) | $(1+2x)^{0.5} =$<br>$1+0.5\times 2x+[0.5\times -0.5\times (2x)^2]/2$<br>$=1+x-\frac{x^2}{2}$ | M1<br>A1 | |
| 6(a)(ii) | $(1-4x)^{-0.5} =$<br>$1+(-0.5)(-4x)+[(-0.5)(-1.5)(4x)^2]/2$<br>$=1+2x+6x^2$ | M1<br>A1 | |
| 6(b)(i) | $\sqrt{f(x)}=(1+x-0.5x^2)(1+2x+6x^2)$<br>$1+3x+7.5x^2$ | M1<br>A1 | |
| 6(b)(ii) | $\lvert x\rvert < 0.25$ | B1 | Accept $-0.25 \le x < 0.25$ |
| 6(c) | $\frac{1+2x}{1-4x}=2$<br>$x=0.1$<br>$2^{0.5}=1+0.3+0.075$<br>$=1.375$ | M1<br>A1<br>A1 | |
| | **Total** | **10** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 7(a) | $[\frac{dy}{dx} =] 3e^{3x} - 24$<br>$[x = 0, \frac{dy}{dx} =] -21$<br>$y = -21x + 1$ | M1<br>A1<br>A1 | |
| 7(b) | $3e^{3x} = 24$<br>$3x = \ln 8$<br>$x = \ln 2$<br>$y = 8 - 24\ln 2$ **ACF** | M1<br>A1<br>A1 | |
| 7(c) | $\frac{d^2y}{dx^2} = 9e^{3x}$ $[= 72]$<br>$[x = \ln 2,]$ $\frac{d^2y}{dx^2} > 0$<br>Hence, min point | B1<br>B1<br>E1 | Must have scored **B1B1** to score this mark |
| | **Total** | **9** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 8(a) | $\left(\frac{dx}{dt}\right)$ rate of change of $x$<br>$(k)$ is proportional to<br>$(80-x)$ amount of substance remaining | E1 | Complete explanation |
| 8(b) | $\int \frac{\mathrm{d}x}{80-x} = \int k\mathrm{d}t$<br>$-\ln(80-x) = kt + c$<br>$t=0,\ x=0,\quad c=-\ln 80$<br>$t=60,\ x=30,\quad -\ln 50 = 60k - \ln 80$<br>$k = \frac{1}{60}\ln 1.6 \qquad [=0.00783...]$ | M1<br>m1<br>A1<br>M1<br>A1 | Separate variables<br><br><br>Attempt to find $k$ |
| 8(c)(i) | $-\ln(80-x) = 2\ln 1.6 - \ln 80$<br>$80 - x = \frac{80}{1.6^2}$<br>$x = 48.75$ | M1<br><br>A1 | |
| 8(c)(ii) | $-\ln(80-70) = \frac{t}{60}\ln 1.6 - \ln 80$<br>$\frac{t}{60} = \frac{\ln 8}{\ln 1.6}$<br>$t = 265$ | M1<br>m1<br>A1 | <br><br>Accept 265 - 266 |
| | **Total** | 11 | |

**Notes: 8(c)** correct answers with **NMS** scores full marks

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 9 | $2y + xy = 1$<br>$x = \dfrac{1-2y}{y}$ | B1 | |
| | $[\text{Vol} =]\ \pi \int_{0.2}^{0.25} (\dfrac{1-2y}{y})^2\ \mathrm{d}y$ | B1 | Correct including $\pi$, limits, $\mathrm{d}y$ |
| | $\int\ = \int y^{-2} + 4 - \dfrac{4}{y}\ [\mathrm{d}y]$ | M1 | Attempt to expand |
| | $= -y^{-1} + 4y - 4\ln y$ | A1 | Correct simplified integral |
| | $= [-4 + 1 - 4\ln 0.25] - [-5 + 0.8 - 4\ln 0.2]$<br>$= 1.2 + 4\ln 0.8$ | m1 | Correct substitution of correct limits into expression in correct form<br>(PI by final answer of 0.964 – 0.966) |
| | $\text{Vol} = \pi(1.2 + 4\ln 0.8)$ **ACF** | A1 | |
| | **Total** | **6** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 10(a) | $x$ / $y$: 1.625 / $1.625^{-1.625} = 0.45432$; 1.875 / $1.875^{-1.875} = 0.30770$; 2.125 / $2.125^{-2.125} = 0.20154$; 2.375 / $2.375^{-2.375} = 0.12817$; 2.625 / $2.625^{-2.625} = 0.07939$; 2.875 / $2.875^{-2.875} = 0.04802$ | B1 | All six correct $x$ values (and no extra used) **PI** by five correct $y$ values |
| | | M1 | At least five correct $y$ values in exact form or decimals, rounded or truncated to three dp or better (in table or formula) (**PI** by AWRT correct answer) |
| | $0.25[0.45432+0.30770+0.20154$ $+0.12817+0.079394+0.04802]$ | m1 | Correct sub into formula with $h=0.25$ **OE** and at least five correct $y$ values either listed, with + signs, or totalled. (**PI** by AWRT correct answer) |
| | $=0.305$ | A1 | **CAO**, must see this value exactly and no error seen |
| 10(b) | $y=x^{-x}$ | | |
| | $\ln y=-x\ln x$ | B1 | |
| | $\frac{1}{y}\frac{dy}{dx}=-x\times\frac{1}{x}-\ln x$ | M1 | |
| | $=-1-\ln x$ | | |
| | $\frac{dy}{dx}=(-1-\ln x)y$ | A1 | |
| | $=(-1-\ln x)x^{-x}$ | A1 | **ACF** |
| | **Total** | **8** | |

**Notes:**
**10(a)** $0.305$ with **NMS** scores **4/4**
**(b)** Correct answer without using implicit differentiation scores **SC2**

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 11(a) | [Equation $\overrightarrow{AB}$] $\mathbf{r} = \begin{pmatrix} 10 \\ 2 \\ -3 \end{pmatrix} + f\begin{pmatrix} -8 \\ -4 \\ 8 \end{pmatrix}$ | M1 | |
| | $= \begin{pmatrix} 10 \\ 2 \\ -3 \end{pmatrix} + \lambda\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ | A1 | |
| | $\lambda = -4f$ | | |
| | $\lambda = -3$ | m1 | |
| | $(4, -1, 3)$ is on line QED | A1 | |
| 11(b)(i) | Coords of $C$ $(4+2c, -1+c, 3-2c)$ | | |
| | $\overrightarrow{DC} = \begin{pmatrix} 6+2c \\ -2+c \\ -4-2c \end{pmatrix}$ oe | M1 | |
| | $\begin{pmatrix} 6+2c \\ -2+c \\ -4-2c \end{pmatrix} . \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = 0$ | m1 | |
| | $12 + 4c - 2 + c + 8 + 4c = 0$ | A1 | |
| | $c = -2$ | m1 | |
| | $C = (0, -3, 7)$ | A1 | |
| 11(b)(ii) | $CD^2 = (0 - -2)^2 + (-3-1)^2 + (7-7)^2$ | M1 | |
| | $= 4 + 16$ | A1 | |
| | $CD = \sqrt{20}$ | | |
| 11(c) | $CP^2 = (4+2p)^2 + (2+p)^2 + (-4-2p)^2$ | M1 | |
| | $= 9p^2 + 36p + 36$ | A1 | |
| | $9p^2 + 36p + 16 = 0$ | M1 | |
| | $p = \dfrac{-36 \pm \sqrt{36^2 - 4 \times 9 \times 16}}{2 \times 9}$ | M1 | |
| | $p = -2 + \frac{1}{3}\sqrt{q}, \qquad p = -2 - \frac{1}{3}\sqrt{q}$ | A1 | oe |
| | **Total** | **16** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 12(a) | $12y\frac{dy}{dx}+8e^{4x}=y^3e^x+e^x3y^2\frac{dy}{dx}$ | M1<br>A1 | Either implicit differential correct |
| | $(12y-e^x3y^2)\frac{dy}{dx}=y^3e^x-8e^{4x}$ | M1 | Or using $\frac{dy}{dx}=0$ |
| | $\frac{dy}{dx}=\frac{y^3e^x-8e^{4x}}{(12y-e^x3y^2)}$ | | |
| | $\frac{dy}{dx}=0,\quad y^3e^x=8e^{4x}$ | A1 | |
| | $q^3=8e^{3p}$ | | |
| | $q=2e^p$ | A1 | **AG** |
| 12(b) | $6\times4e^{2p}+2e^{4p}=8e^{3p}e^p$ | M1 | Equation all in terms of one variable |
| | $24e^{2p}=6e^{4p}$ | A1 | |
| | $e^{2p}=4$ | m1 | |
| | $p=\ln2$ | | |
| | $q=4$ | A1 | ACF |
| | **Total** | **9** | |

| Q | Answer | Marks | Comments |
|---|---|---|---|
| 13 | $\frac{dx}{dt} = 2at$<br>$\frac{dy}{dt} = 2a$<br>$\frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$ | M1 | Either differential correct |
| | $At\ P,\ y - 2ap = \frac{1}{p}(x - ap^2)$ | m1 | Equation of a tangent with $t$ replaced |
| | $yp = x + ap^2$<br>$yq = x + aq^2$ | A1 | Both correct (unsimplified) |
| | $At\ R,\ y(p-q) = a(p^2 - q^2)$ | m1 | Attempt to solve |
| | $y = a(p+q)$<br>$x = apq$ | A1 | Both correct |
| | $y^2 = a^2(p^2 + q^2 + \frac{2x}{a})$<br>$= a(ap^2 + aq^2 + 2x)$ | m1 | |
| | $p^2 + q^2 = 1,$<br>$y^2 = a(a + 2x)$ | A1 | |
| | **Total** | **7** | |